STRATUM

01 · CONCEPT *Move the plane. Lose the cloud.*

CHALLENGE	Conrad Challenge 2026–27
CATEGORY	Aerospace & Aviation
CHAPTER	Global
STAGE	Activation · Lean Canvas
DATE	12 Oct 2026

02 PROBLEM

What need are we addressing?

Airlines face rising climate pressure but can only cut carbon, which is slow and costly. Contrails cause about half their warming and are far cheaper to avoid, yet no airline can tell which flights will make them.

04 SOLUTION

Key characteristics

Stratum combines public flight tracks with weather data to predict where contrails will persist, then corrects unreliable humidity forecasts using satellite-observed contrails. Output: a ranked list of flights, the altitude change needed, and the warming-versus-fuel tradeoff for each.

05 UNIQUE VALUE PROPOSITION

Why us over everything else

Existing tools trust raw humidity forecasts, which are wrong often enough that airlines won't act on them. Stratum learns from satellite-observed contrails to correct those forecasts, and shows the fuel cost against warming avoided, built for Indian airspace first.

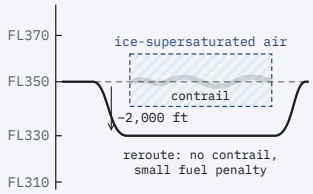


Fig. Most avoidance is a small altitude change around a thin layer of ice-supersaturated air. Illustrative, not to scale.

06 SUSTAINABLE ADVANTAGE

Why it's hard to copy

Every flight we evaluate and every satellite image we check adds to a validation dataset for Asian skies that sharpens the model each time it runs. Early airline partnerships and regulator relationships compound that lead.

07 CUSTOMER SEGMENTS

Who we serve

Airline flight dispatch and sustainability teams, starting with Indian carriers like IndiGo and Air India, who face climate targets and need cheap, verifiable reductions. Air navigation providers are the second segment.

03 EXISTING ALTERNATIVES

How is it solved today?

Most airlines ignore contrails and spend on offsets or sustainable fuel instead. Contrail avoidance exists only in small trials, Google with American Airlines, Breakthrough Energy's tools, over US and European airspace, not Indian.

10 KEY METRICS

Numbers that track success

Accuracy: hit rate and false alarm rate against satellite contrail observations. Impact: net warming avoided after accounting for extra fuel. Adoption: flights screened monthly, rising to paid airline subscriptions. Accuracy gates everything, a false alarm wastes fuel for nothing.

09 CHANNELS

How we reach and deliver

We give the map away. A free web dashboard showing contrails over Indian airspace earns citations and press; airlines come to us. Later, an API pushes the same alerts into the flight-planning software dispatchers already use.

08 EARLY ADOPTERS

Our very first users

Atmospheric researchers and climate NGOs studying Asian aviation, who need contrail data that doesn't exist for this region and will test unproven tools. They cost nothing to serve, give expert feedback, and their validation is what makes airlines listen later.

11 COST STRUCTURE

Compute dominates: running contrail physics and model inference across thousands of flights daily, plus storing the validation dataset. Flight and weather data are free and open. Costs scale with airspace covered, not customers, so margins improve with scale.

12 REVENUE STREAMS

Airlines pay an annual subscription per aircraft for contrail alerts. Verified avoidance sold as carbon credits, priced per tonne of warming avoided, with us taking a share. Early on, climate and aviation research grants fund development while we validate.

+ ADDITIONAL POINTS

The physics we build on

Schmidt–Appleman criterion: does a contrail form? Ice supersaturation (`RHi > 100%`): does it persist? Both are textbook. The weak link is the humidity forecast at cruise altitude.

A few flights do most of the damage

A study over Japan found about 2% of flights caused 80% of contrail warming. We don't need to move every plane, just the right ones.

Night flights matter most

By day a contrail also reflects some sunlight. At night it only traps heat, so overnight routes (mostly cargo) are where avoidance pays off most.

Open data only

OpenSky (ADS-B flight tracks), ERA5 (ECMWF weather reanalysis), INSAT-3D / Himawari satellite imagery for validation. Anyone can check our work.

Assumptions we still have to test

1. Satellites can spot contrails through monsoon cloud cover.
2. Corrected forecasts beat raw ERA5 humidity.
3. Dispatchers will act on an alert.

Biggest risk

Contrail warming estimates carry real scientific uncertainty, so we report ranges, not single numbers.

Next steps

Find an atmospheric-science advisor (IISc CAOS). Run the formation check on one month of Delhi–Mumbai traffic.

Refs: Lee et al. 2021, Atmos. Environ. (aviation climate forcing) · Teoh et al. 2020, Environ. Sci. Technol. (contrail mitigation over Japan)